

The Language Areas

The human brain, unlike that of any other animal, has areas dedicated specifically to language, usually located in its left hemisphere. The unique ability of humans to communicate using language is thought to be an evolutionary advantage.

Broca's and Wernicke's areas

The two main language areas are Broca's and Wernicke's areas. Broca's area is associated with moving the mouth to articulate words. When learning new languages, separate parts of Broca's area are activated when we speak either our native or non-native tongue. In Wernicke's area, words that we hear or read are understood and selected for articulation as speech. Damage to this part of the brain can lead people to speak in peculiar ways, creating sentences that do not make sense.

Motor cortex

The motor cortex enables the physical movements required to produce language—for example, moving your tongue, lips, and jaw. The motor cortex is activated when words that are semantically related to body parts are heard or spoken. For example, the word “dance” might be related to your feet.

Speech travels through air as sound waves

BRAIN DAMAGE AND LANGUAGE CHANGES

There have been cases in which patients with brain injury appeared to wake up speaking a different language or with a different accent. Foreign accent syndrome is one example of such a medical condition. These cases are rare, and there have not been sufficient scientific studies carried out to understand them in any detail.



& @ å
ž ø ï ¿ œ
» § ë

HELLO

SHWMAE BONJOUR

ASALAAM ALAIKUM

GUTEN TAG

PRIVET OLÁ

KONNICHIIWA

HOLA CIAO

Speaking and understanding

Processing language is a complex task. Articulating or decoding even a simple greeting, such as “hello,” requires several different areas of the brain to work together.